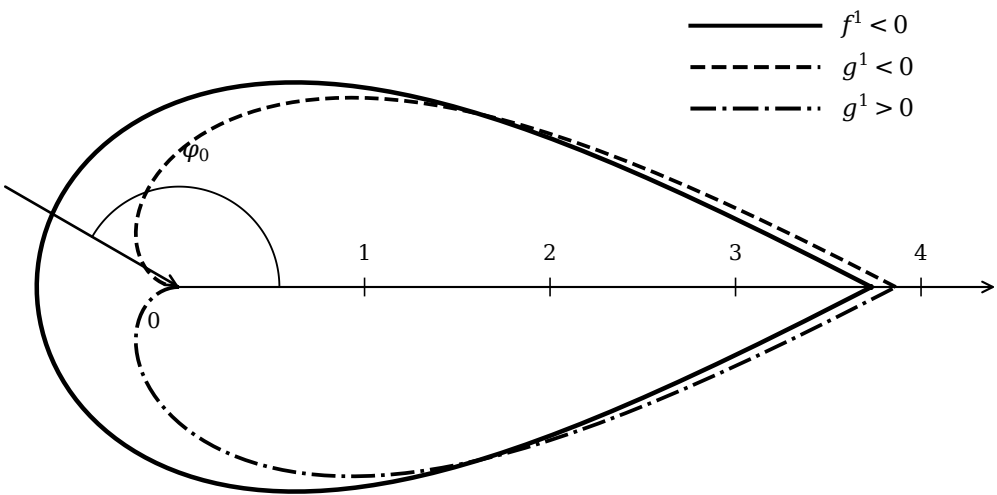

$\varphi_0 = 150^\circ, \quad \alpha = 360^\circ$